

4. Group Project Report (Group: 40%)

Assessment 4: Group Project Report

Due Date: refer to Learnline

Weighting: 40%

Group Size: 4 students (same group as in Assessments 1 and 2)

Learning Outcomes Assessed

- Unit Learning Outcome 1: Apply appropriate statistical techniques to evaluate data analysis models.
- Unit Learning Outcome 2: Critically analyse the role of data analysis task in an organisational context.
- Unit Learning Outcome 3: Justify the process or methods used to analyse large data sets.
- Unit Learning Outcome 4: Design and implement a pipeline to analyse a large heterogenous dataset.
- Unit Learning Outcome 5: Ability to process large data sets using appropriate technologies.
- Unit Learning Outcome 6: Apply communication skills to present data processing results, orally and in written form.

Task Description

This assessment requires you to delve deeper into avian disease surveillance data, expand upon your previous work, and apply advanced classification techniques. You will construct a robust dataset, perform comprehensive data preprocessing, conduct exploratory data analysis through visualisations, implement and evaluate classification models, and contextualise your findings within the broader impact on Great Britain.

This assignment directly addresses the learning outcomes as follows:

- *Apply appropriate statistical techniques to evaluate data analysis models.*

You will select and apply classification models, evaluate their performance using appropriate metrics, and interpret the results to draw meaningful conclusions.

- *Critically analyse the role of data analysis tasks in an organisational context.*

You will contextualise your findings within the broader implications of avian disease surveillance, demonstrating how data analysis supports decision-making in an organisation (such as government agencies, agricultural businesses, conservation organisations, etc.)

- *Justify the process or methods used to analyse large datasets.*

You will justify your choice of preprocessing techniques, classification models, and evaluation metrics, ensuring alignment with the problem context and dataset characteristics.

- *Design and implement a pipeline to analyse a large heterogeneous dataset.*

You will design and implement a data analysis pipeline, including data preprocessing, exploratory data analysis, model training, and evaluation.

- *Ability to process large datasets using appropriate technologies.*

You will demonstrate proficiency in using Python and relevant libraries (e.g., Pandas, Scikit-learn, Matplotlib, Seaborn, and/or other relevant packages) to process and analyse data.

- *Apply communication skills to present data processing results, orally and in written form.*

You will communicate your findings effectively through a professionally written report tailored to potentially technical and non-technical audiences.

Context Statement

(please refer to Assessment 1)

Step-by-Step Instructions

1 - Dataset Creation and Preprocessing:

- Substantially extend the existing dataset or create a new dataset from the Tableau avian dashboard (<https://public.tableau.com/app/profile/siu.apha/viz/AvianDashboard/Overview> (<https://public.tableau.com/app/profile/siu.apha/viz/AvianDashboard/Overview>)) provided at the start of Assessment 1, including other relevant sources if planned earlier.
- Preprocess the dataset appropriately for a classification task. This includes steps such as handling missing data, encoding categorical variables, scaling features, and splitting the data into training and testing sets.
- Document the data sources, data collection methods, and any data integration steps.
- Provide a detailed explanation of the preprocessing steps taken and why they were chosen.

Skills acquired: These tasks address LO4 (pipeline design) and LO5 (appropriate technologies) by requiring systematic data gathering and integration. They also emphasise LO3 (justification of methods) by requiring reasoned choices and acknowledging limitations.

2 - Exploratory Data Analysis (EDA):

- Generate effective data visualisations to explore the dataset, identify patterns, and inform the classification task.
- Use visualisations to communicate key insights about the dataset, such as class distributions, correlations, and potential outliers.

Skills acquired: These tasks strengthen the statistical aspect of LO1 (statistical techniques).

3 - Classification Modelling:

- Run at least **two (2)** classification models that have been taught in this unit to address **one (1)** classification problem (for example, predicting disease outbreaks, identifying high-risk regions, classifying disease types, or another relevant classification task).
- It is desirable that the formulated classification task be informed by the outcome of regression modelling done in Assessment 2.
- Optimise the models using techniques such as hyperparameter tuning, cross-validation, or feature selection.

Skills acquired: These tasks assess LO1 (statistical techniques) and LO5 (appropriate technologies) by requiring statistical optimisation and coding proficiency.

4- Model Evaluation and Discussion:

- Evaluate and compare the performance of the models using appropriate metrics (e.g., accuracy, precision, recall, F1-score, ROC-AUC). Justify the choice of metrics based on the problem context.
- Provide specific recommendations for a selected organisation based on the findings.

Skills acquired: These tasks reinforce LO1 (statistical techniques) and LO3 (justification of methods) by requiring you to report statistical evaluation and provide interpretation.

5 - Professional Report Writing:

Document the entire process in a professionally written report, including:

- A clear description of the dataset, preprocessing steps, and classification problem.
- Justification of the chosen methods and models (refer to task descriptions)
- Interpretation of the results and their implications for avian disease surveillance.
- A discussion of how the findings could inform decision-making for the organisation in focus.
- **Executive Summary** tailored to a non-technical audience.
- A **flowchart or diagram** describing your data analysis pipeline.

Skills acquired: These tasks assess LO2 (data analysis in an organisational context) and LO6 (written communication skills).

Deliverables

A Group Report (maximum 2,000 words per student) **with tracked changes**:

- Professionally written, detailing the dataset, methodology, results and discussion.
- Include visualisations, tables, and references to support the analysis.

A copy of Turnitin originality report.

A copy of the signed AI Usage Declaration Form:

- Include clear acknowledgement of the use of generative AI.
- A reflective account of the use of generative AI in the project, accompanied by well-documented prompts and interaction evidence.
- Please use the following template:



AI Usage Declaration Form.docx

Python code:

- Well-documented Python code used for preprocessing, visualisation, and modelling.
- Include comments explaining key steps and decisions made regarding code designs.

Submission Guidelines

Please submit one (1) .zip file containing:

- A group report in Microsoft Word format (.docx) **with tracked changes**. Incorporating a professional title page and table of contents is encouraged; they do not contribute to the word count.
- A copy of Turnitin originality report.
- A copy of AI Usage Declaration Form signed by all team members.
- A text file (.txt) containing a link to Python codes on GitHub or BitBucket.

Submit your work via Learnline *no later than* the stated due date of this assignment.

- Email submissions will not be accepted.
- Do not submit multiple documents.
- Late submission penalty may apply: <https://policies.cdu.edu.au/view-current.php?id=177#major4> (<https://policies.cdu.edu.au/view-current.php?id=177#major4>)

Resources

- Publicly accessible Tableau avian dashboard:
<https://public.tableau.com/app/profile/siu.apha/viz/AvianDashboard/Overview>
(<https://public.tableau.com/app/profile/siu.apha/viz/AvianDashboard/Overview>)

Overall Assessment Criteria

The assessment will be based on the following criteria:

- Ability to construct a high-quality dataset and preprocess it effectively for a classification task.
- Creation of insightful and effective data visualisations.
- Competence in running and optimising classification models, and justifying the chosen methods and models.
- Critical evaluation of model performance using appropriate metrics.
- Professional writing style, clarity, and organisation of the report, and the ability to contextualise findings for a non-technical audience.
- Responsible, effective and appropriate use of AI tools.

Originality

Do not plagiarise from other student groups.

Do not engage in contract cheating. Providing contract cheating services is a criminal offence in Australia, punishable by imprisonment and/or hefty fines. Reference:
<https://universitiesaustralia.edu.au/media-item/cheating-the-system/>
(<https://universitiesaustralia.edu.au/media-item/cheating-the-system/>).

You may choose to use generative AI tools (e.g., ChatGPT, Gemini, DeepSeek or others) to brainstorm or assist with generating non-critical elements of your Python code. You must demonstrate a clear understanding of all elements of your code and their functionality by including explanations of critical code sections in your report.

In writing reports, you **must not** use AI tools to write and conceptualise the first draft of your report. This is an important stage where students are expected to clearly demonstrate their original thoughts and ideas. You may use AI tools to improve the clarity of your writing at a later stage and/or to supply non-critical details and background information. If so, these must be clearly acknowledged in your AI Usage Declaration Form that is signed by all.

The group mark will not be released in the absence of the mandatory AI Usage Declaration Form.

Violating the originality requirement may constitute a serious breach of academic integrity, leading to significant loss of marks or even not passing the unit.

Individual Grades

This unit is committed to ensuring that each team member contributes his or her fair share of the group work.

All student have the responsibility to clearly document their contributions. Failure to provide satisfactory evidence of individual contributions may result in loss of marks or not passing this assessment for the student involved.

All teams **must** document their creative process on Microsoft Teams and GitHub/BitBucket. Other platforms (WhatsApp, Facebook, Discord, etc.) **should not** be used as evidence, without any prior agreement with the campus lecturer. The group mark may not be released to the students if this collaborative requirement is not met by the submission due date.

Evidence from Microsoft Teams and GitHub/BitBucket may be used to further differentiate individual marks. Your instructor reserves the right to differentiate individual grades where appropriate.

If in doubt, please contact your campus lecturer.

Academic Integrity and Assessment Irregularities

Academic integrity is a core value at CDU and must be upheld at all times when completing this assignment. You must not plagiarise the work of others. Please refer to the [Students - Breach of Academic Integrity Procedures \(https://policies.cdu.edu.au/view-current.php?id=50\)](https://policies.cdu.edu.au/view-current.php?id=50).

Other assessment irregularities are governed by CDU's Higher Education Assessment Procedures (<https://policies.cdu.edu.au/view-current.php?id=177>).

Acknowledgement of the use of generative AI

I, Yakub Sebastian, declare that I originally drafted the critical design elements of this assessment: ideation, contextualisation, Tableau dashboard selection and the outline of assessed tasks.

I acknowledge the use of ChatGPT, DeepSeek and Gemini in further articulating the detailed wording of the assessments based on prompts I originally crafted. These prompts are well-documented and may be referred to upon request.

I further declare that I used these AI tools to generate the initial grading rubric by providing the assessment details and the overall assessment criteria. I then carefully vetted all AI-generated outputs and modified them in accordance with my expert knowledge to ensure that all assessment tasks and the rubrics evaluate all designated unit learning outcomes at appropriate cognitive levels.